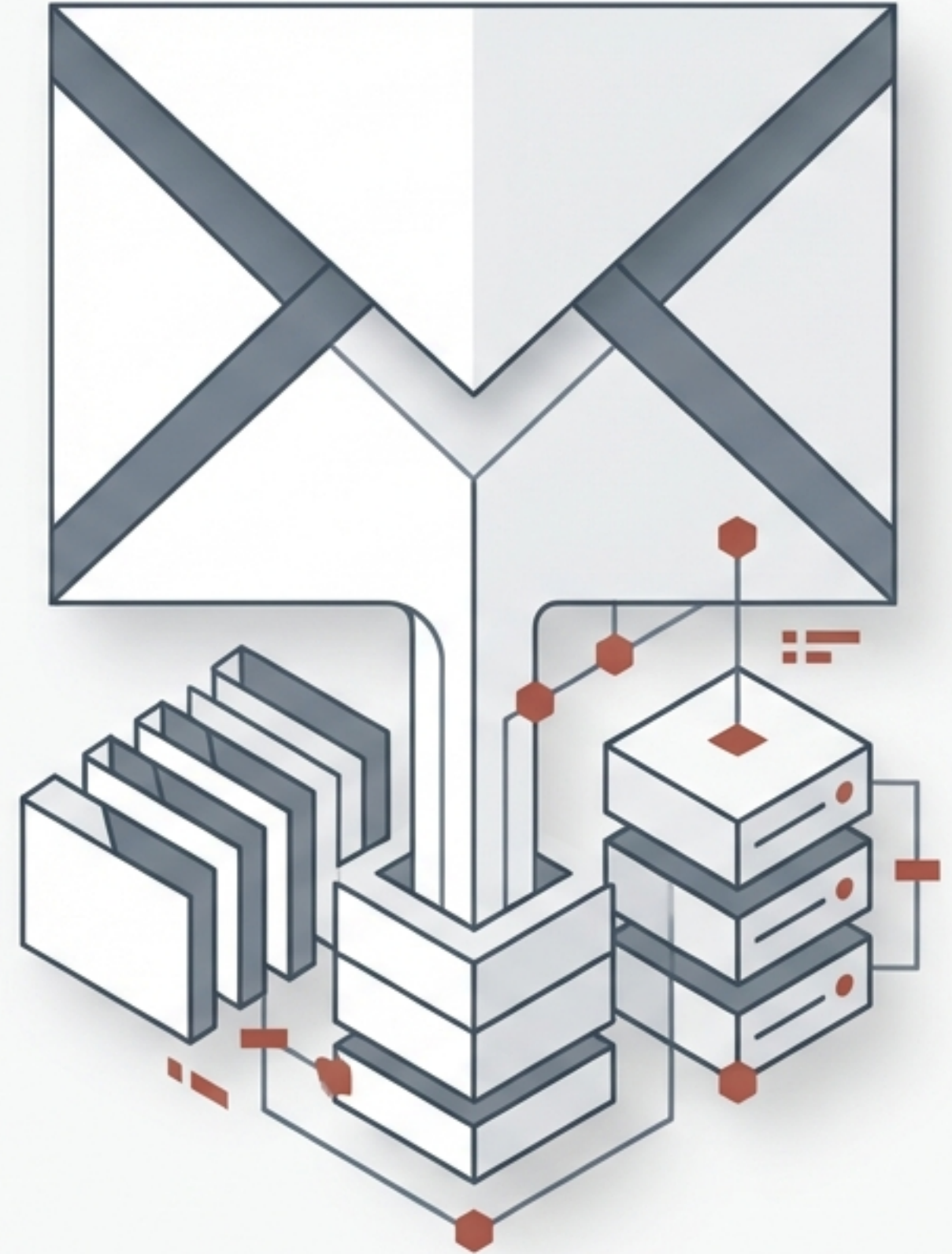


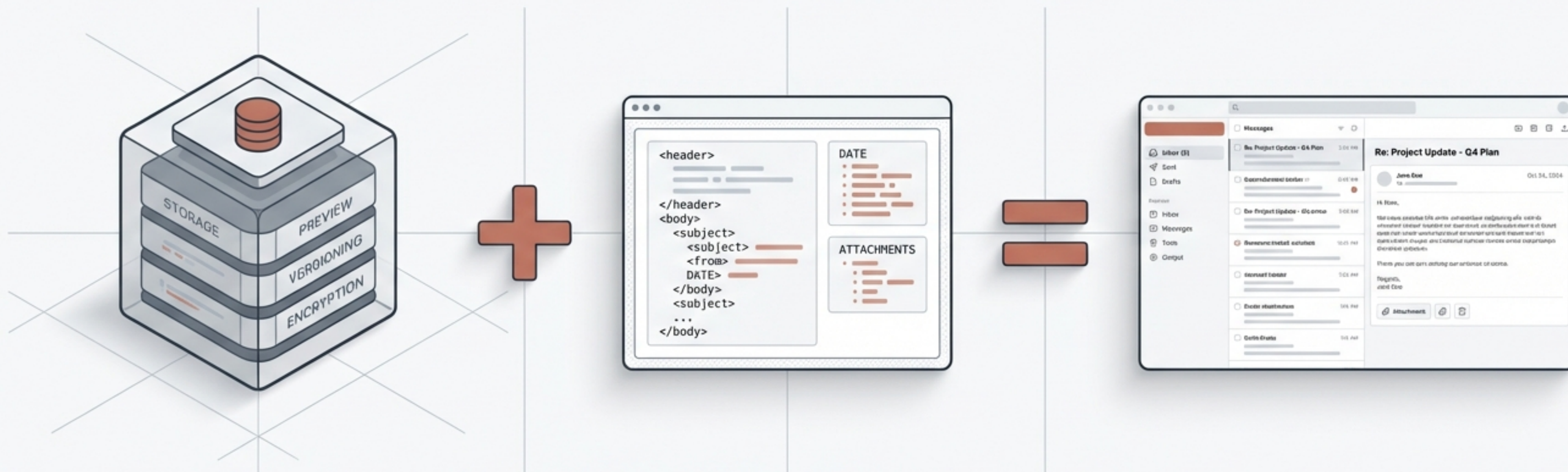
Arch Brief v0.7.4: Email-FS

- Date: 27 Feb 2026
- From: Human (project lead)
- To: Architect (lead), Developer, Librarian
- Type: Arch brief — new primitive, design phase



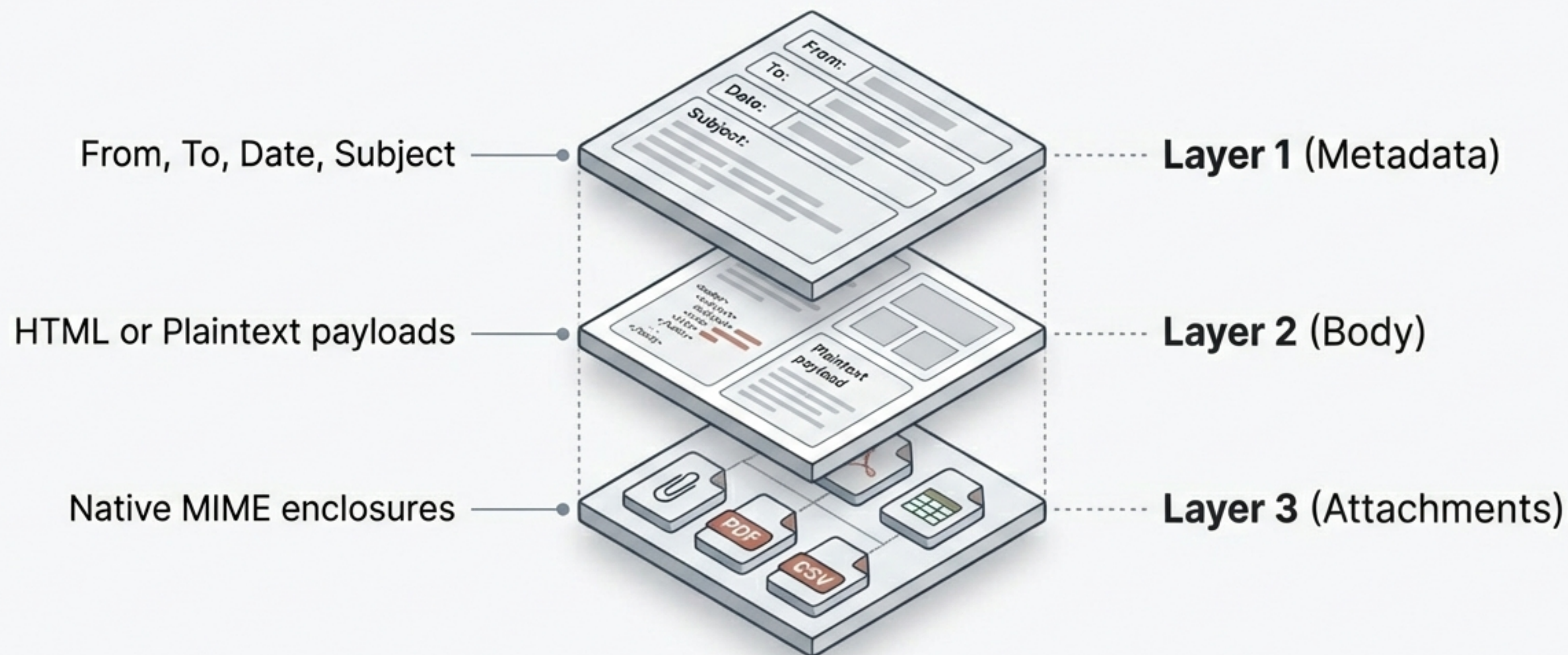
The 90% Rule for Building an Inbox

Data Room + EML Renderer = Functional Inbox



- We already possess **90% of a mini email client**.
- If the existing data room can render EML files exactly as it renders PDFs and CSVs, we instantly gain a **functional inbox** requiring zero new backend infrastructure.

Demystifying the RFC 5322 Envelope



Takeaway: An EML file is a self-contained, well-defined envelope. The preview component treats it as another natively supported format alongside PDF, CSV, and Markdown.

Folders Automatically Become Functional Inboxes

Project Documents - Emails	
	20240901_MeetingNotes.eml
	20240901_Q4_Roadmap.eml
	20240831_UrgentUpdate.eml
	20240830_ClientFeedback.eml
	20240830_InvoiceRequest.eml

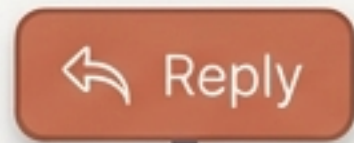
PREVIEW COMPONENT
TRANSFORMATION

Inbox - Project Documents		
	Jane Doe	Meeting Notes Sep 1, 2024
	John Smith	Q4 Roadmap Discussion Sep 1, 2024
	Alex Chen	URGENT: Update Required Aug 31, 2024
	Maria Garcia	Client Feedback on Draft Aug 30, 2024
	Finance Dept	Invoice Request - August Aug 30, 2024

- The messages are standard files.
- The inbox is a standard folder.
- Sorting a folder of EML files by date instantly produces an inbox view.
- Security is inherently handled by existing data room encryption.

Composing and Replying Natively

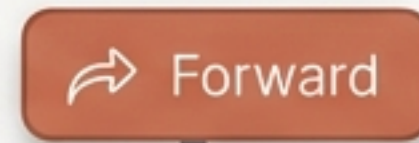
Reply



Generates a new EML file containing an In-Reply-To header referencing the original.

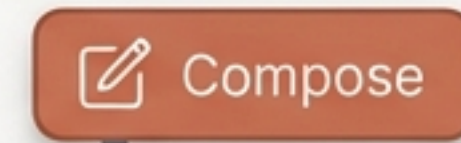
Threads rely purely on standard email header chaining.

Forward



Generates a new EML packaging the original body and attachments with a blank To: field.

Compose

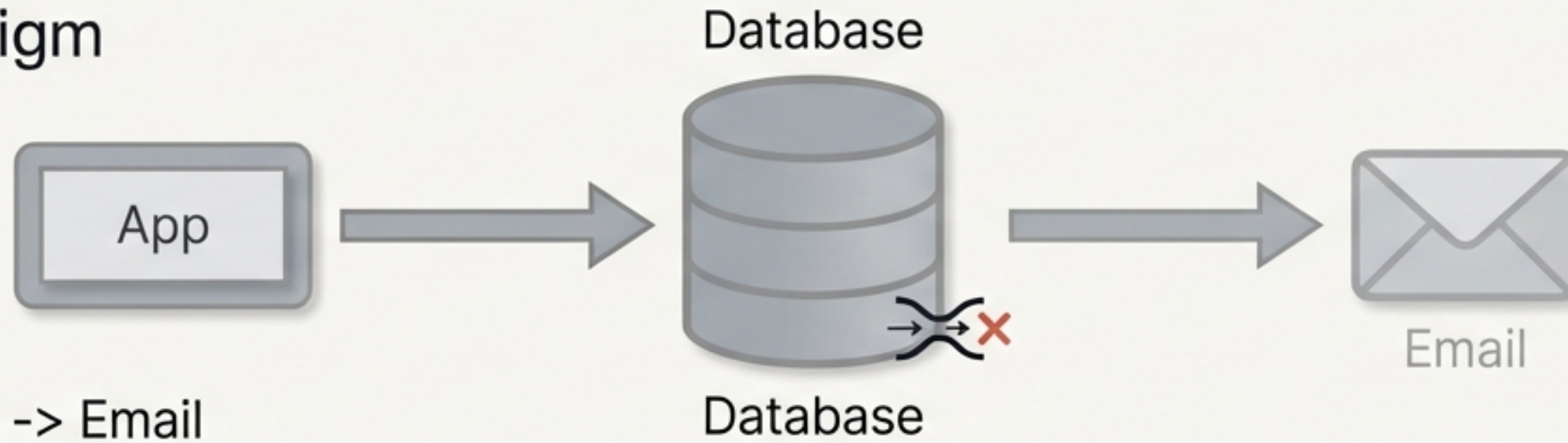


Generates a raw EML from scratch, saving directly into the data room.

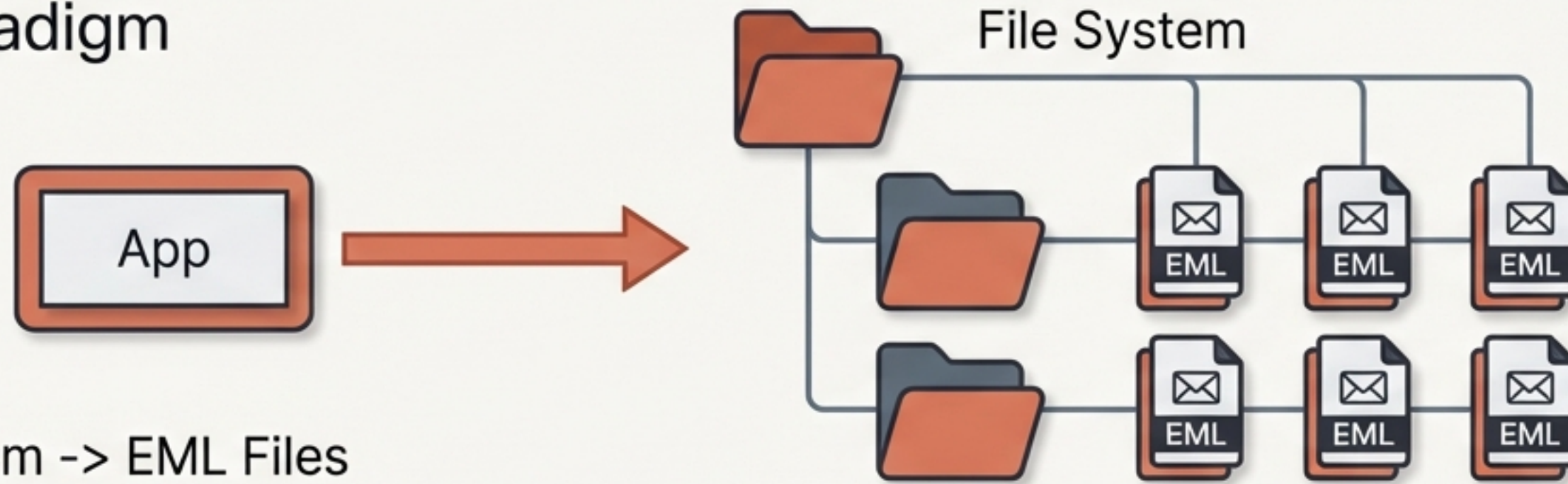
Sends externally via SES workflows.

The File System Functions as the Email Store

Legacy Paradigm



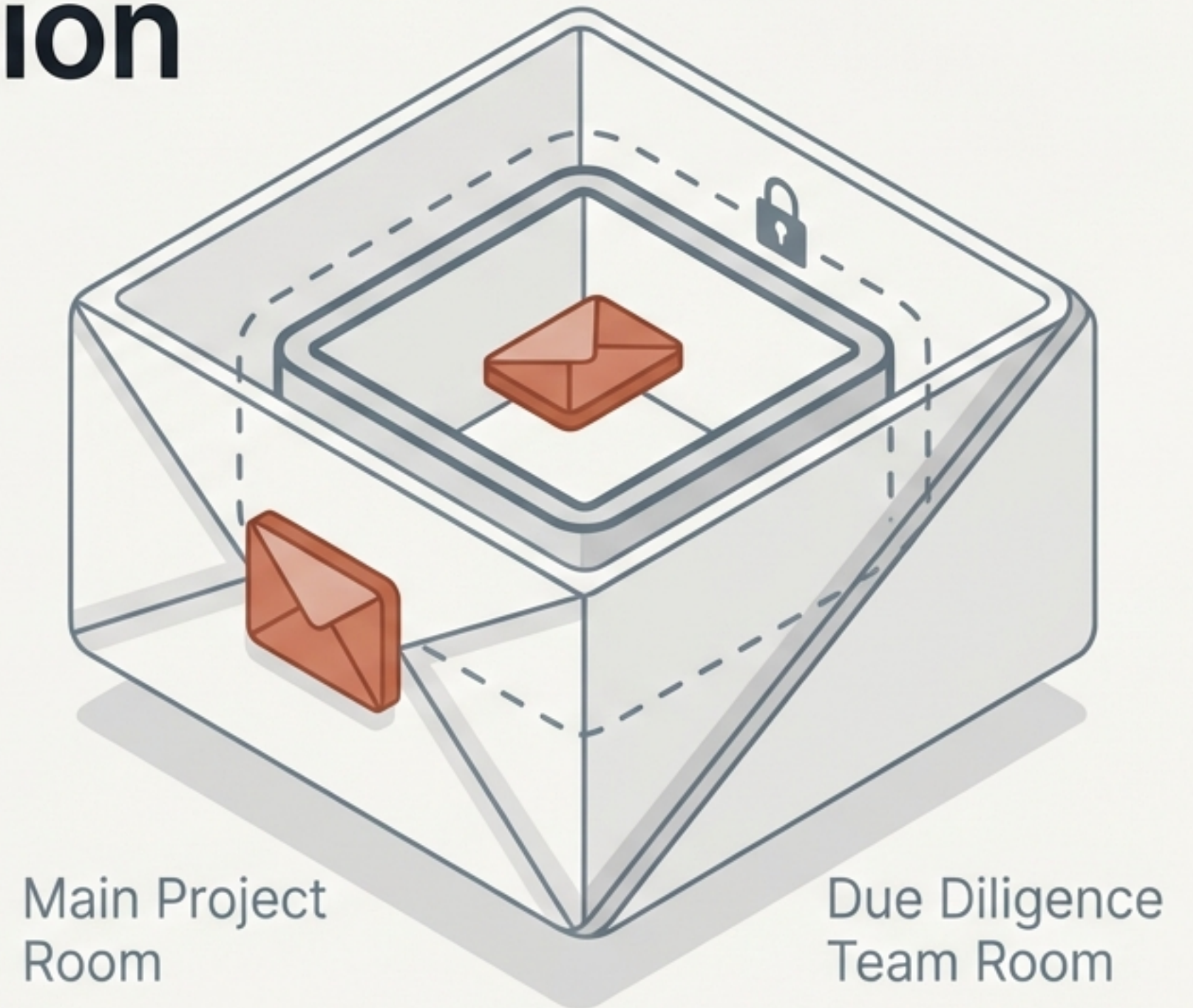
Email-FS Paradigm



Bypassing traditional database bottlenecks. Issues-FS utilises the file system as a database for task tracking; Email-FS applies the exact same principle to messages. Every data room natively houses its own email system.

Fractal Architecture Enables Compartmentalisation

- Data rooms within data rooms contain independent email systems.
- Parent systems cannot see child systems unless explicitly shared.
- Ensures strict compartmentalisation for sensitive workflows.



Email as a Primitive for AI Agents

Email is no longer just a user feature. It is a highly structured communication **primitive** for agents.

By forcing agents to communicate via email primitives stored in **data rooms**, we secure perfect **traceability**, **verifiability**, and a universally familiar interface for human inspection.

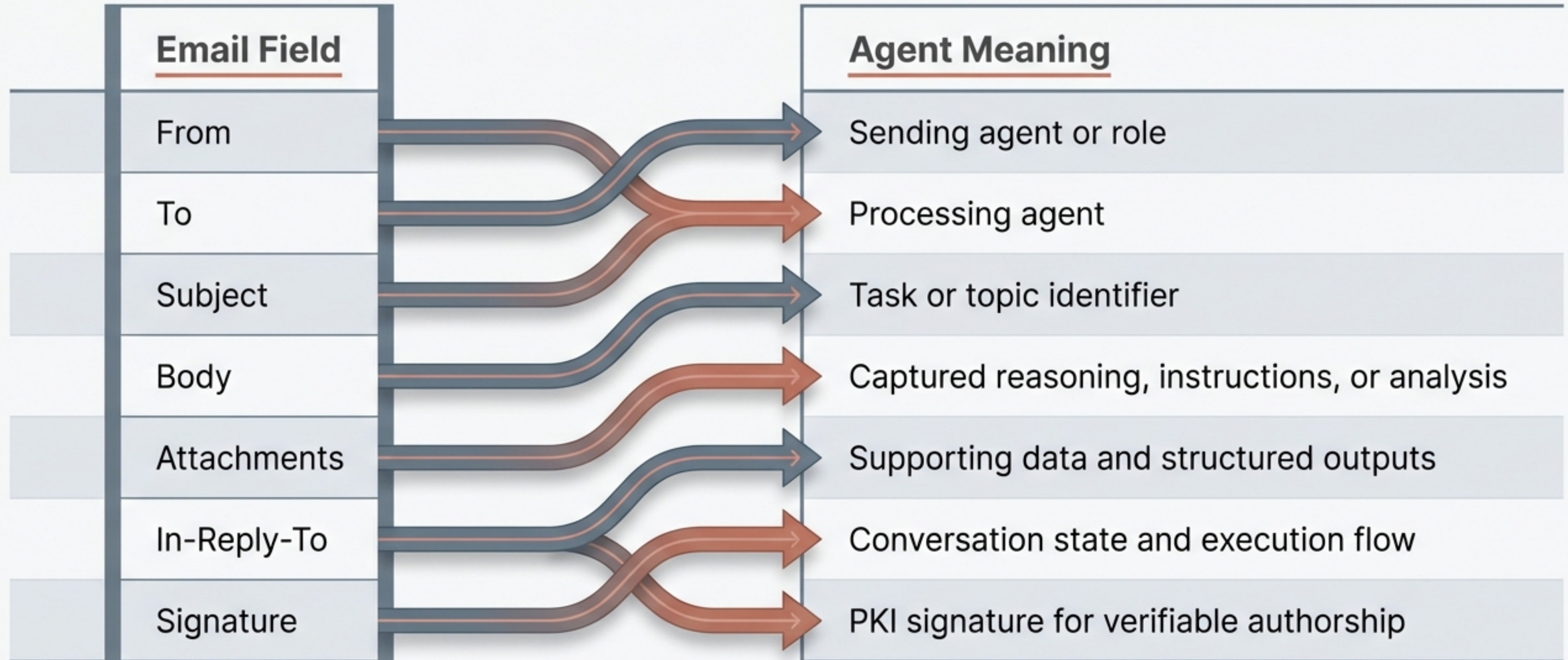


Legacy Human
Communication

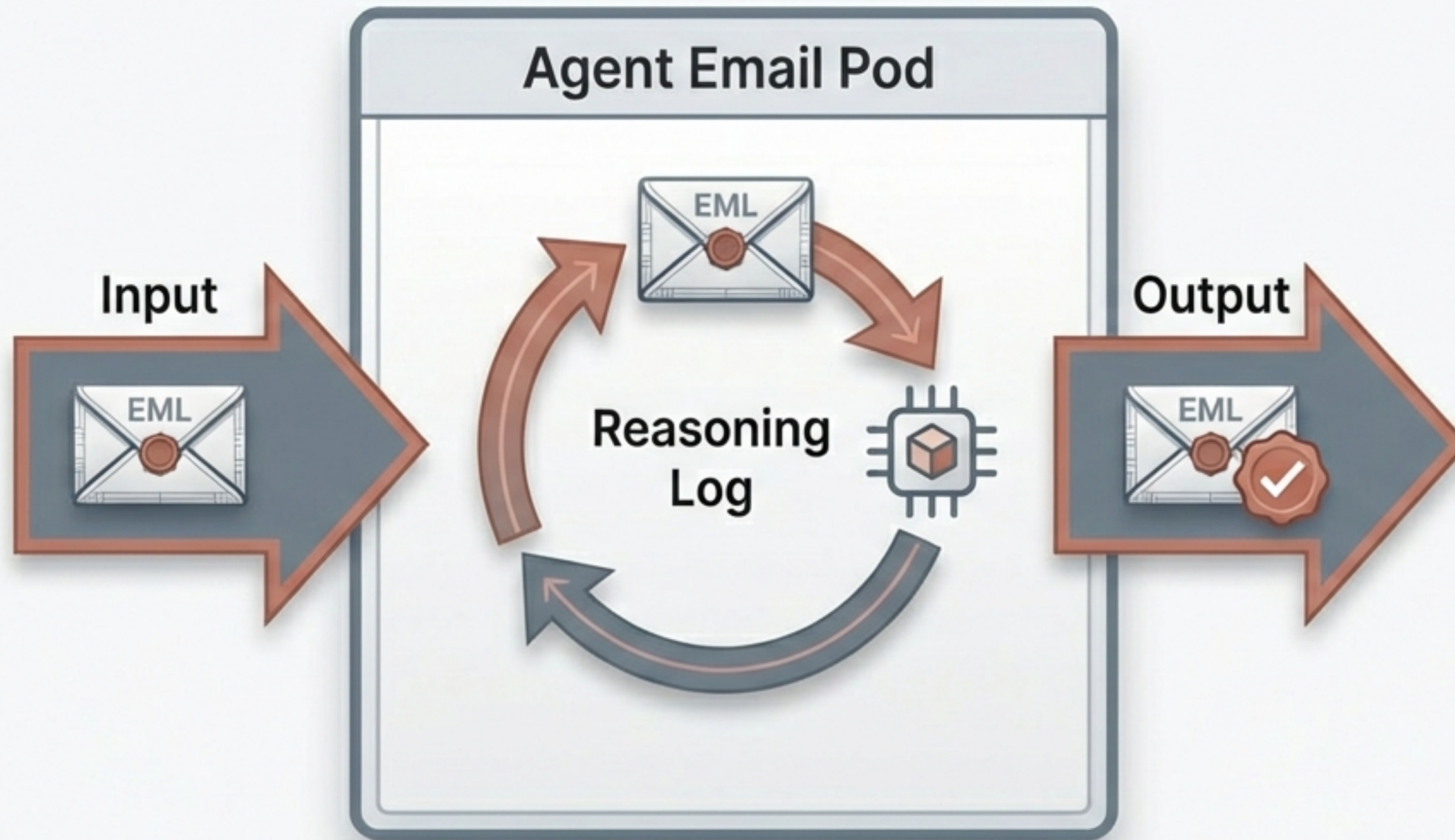


Machine
Communication

Mapping Standard Protocols to Agent Workflows

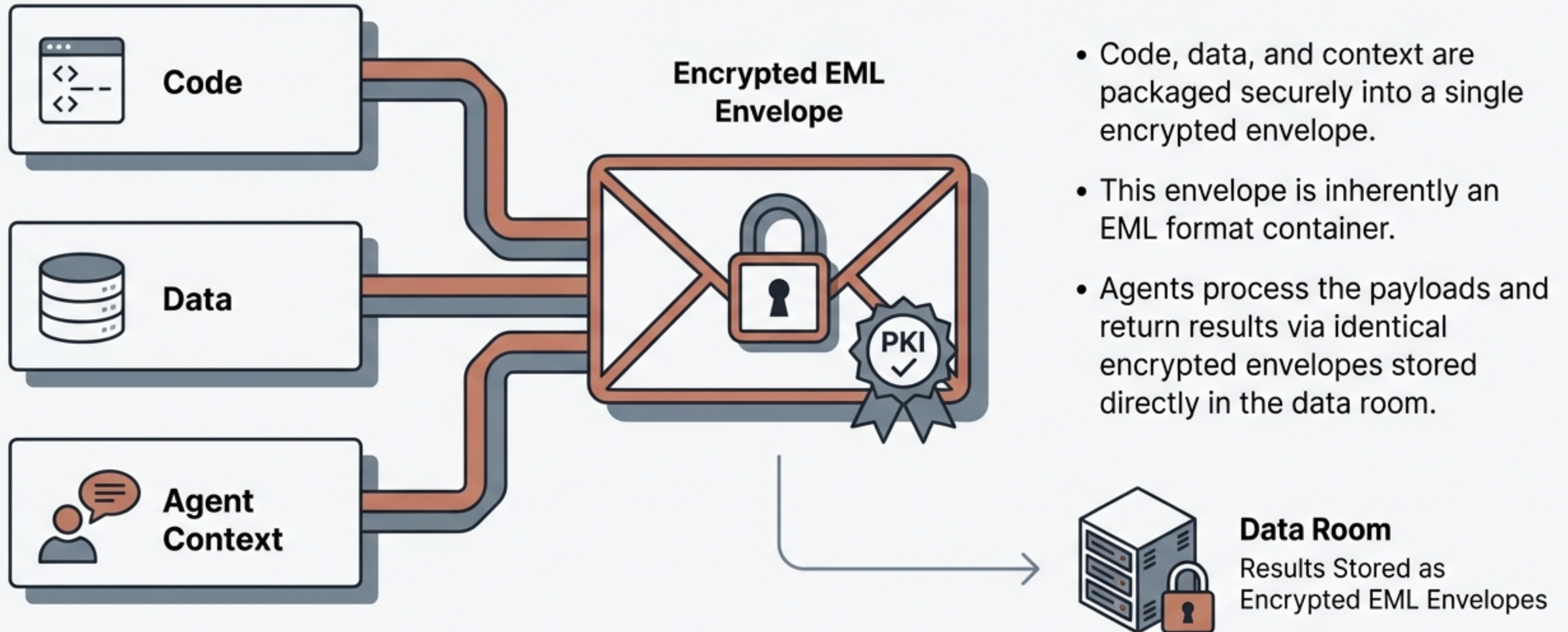


Execution Environments Run on Email Contexts



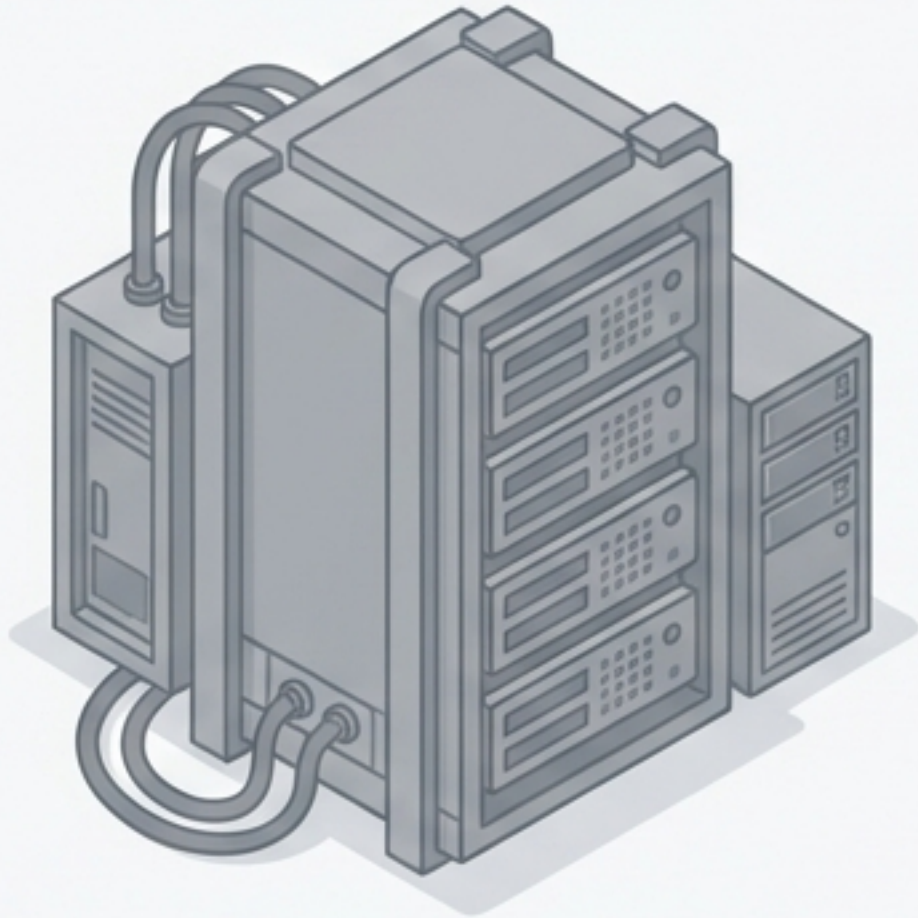
- The pod receives task inputs entirely as email.
- The agent logs internal reasoning as self-addressed emails, creating a permanent, human-inspectable record of autonomous decisions.
- All outputs are formatted as verifiable, signed EML files.

Securing the Agent Supply Chain with EML Envelopes



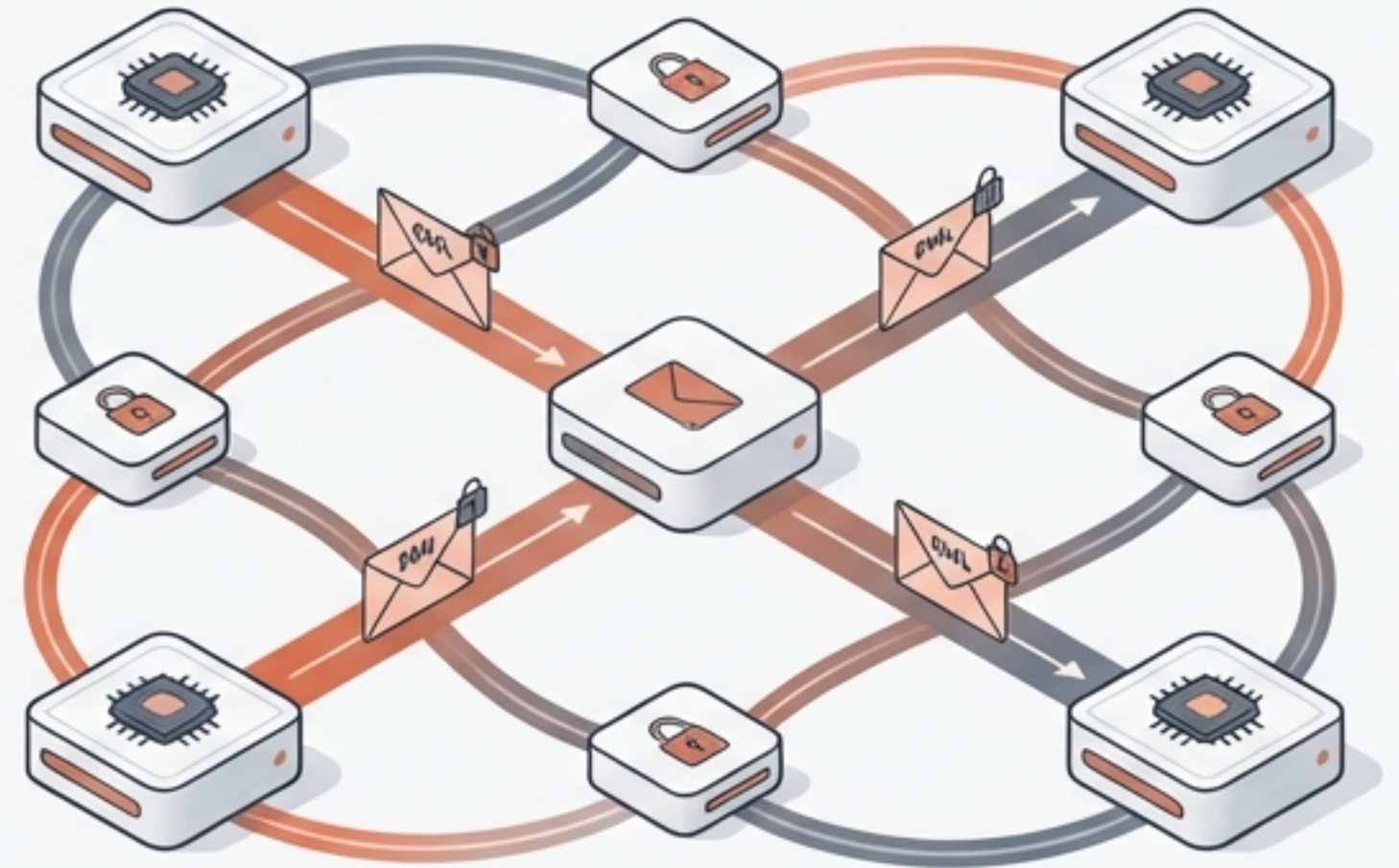
Modernising Beyond Frozen Enterprise Architectures

Legacy Inertia



2010 Enterprise Server

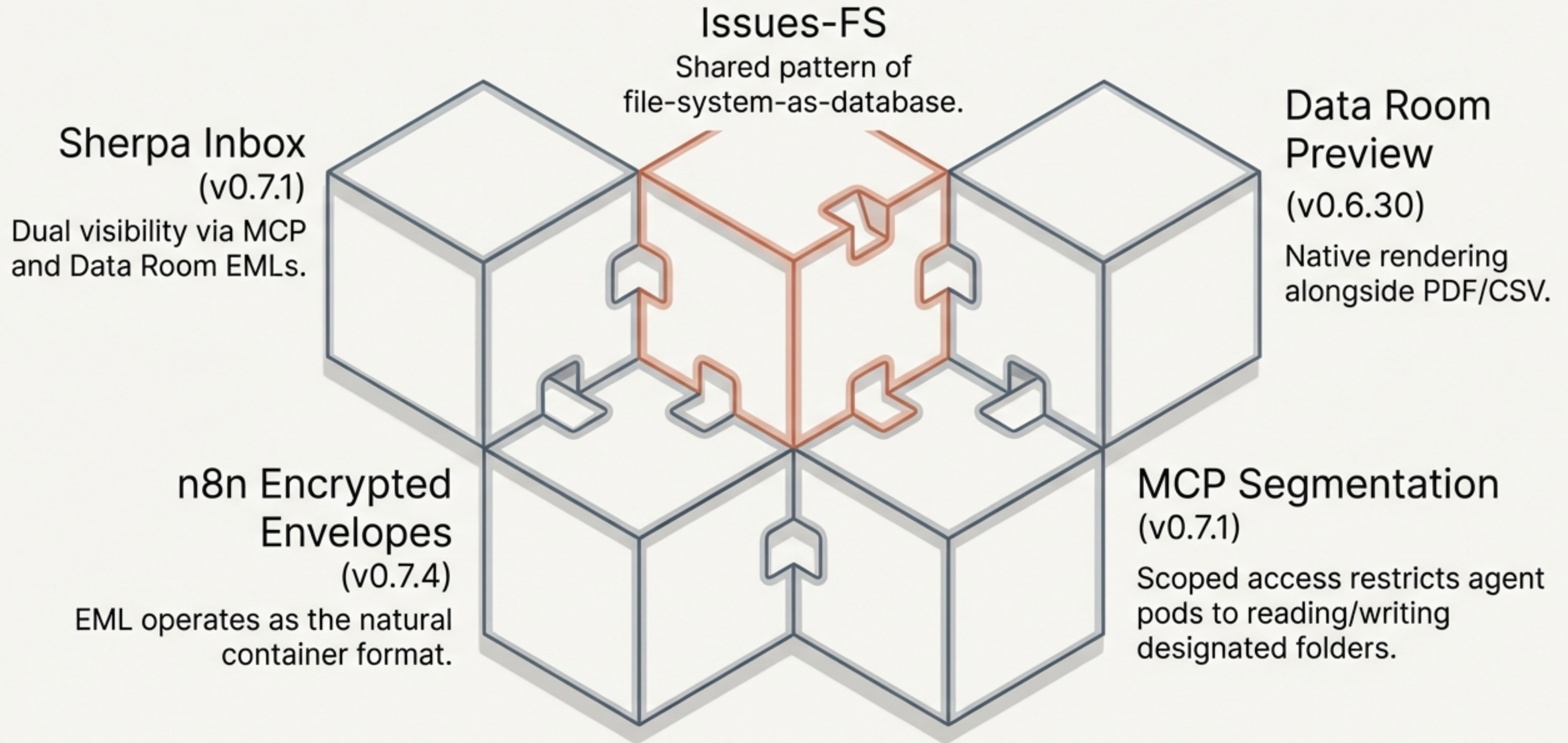
Modern Ecosystem



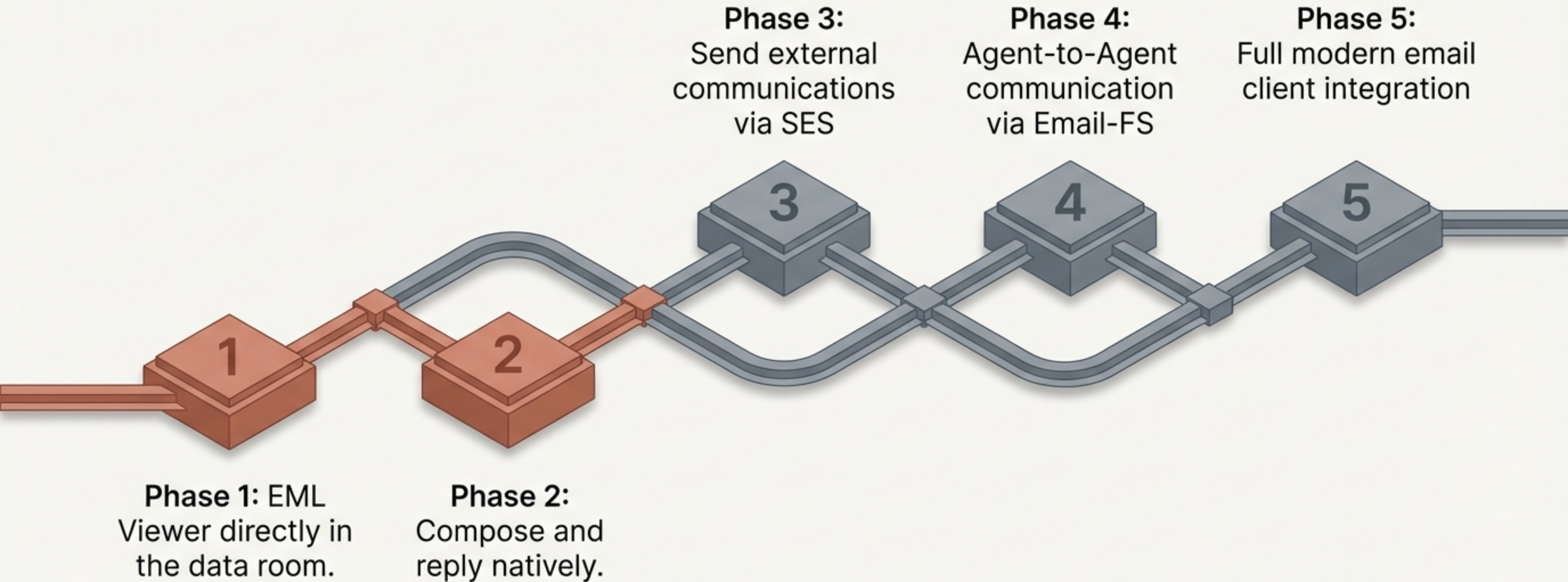
Agent pods

- Legacy enterprise email systems are maintained, not improved. UIs are dated, and APIs remain restricted.
- We are not building a day-one email replacement; we are demonstrating a modern, encrypted, agent-native ecosystem built incrementally inside the data room.

Locking Email-FS into the Existing Ecosystem



Incremental Path to an Agent-Native Client



Note: Every single phase delivers independent, immediately deployable utility.

Acceptance Criteria for Immediate Sprints

Phase 1 (EML Viewer)

- ✓ • Renders sanitised HTML/plain text bodies.
- ✓ • Displays From, To, CC, Subject, Date.
- ✓ • Lists attachments with download buttons.
- ✓ • Automatically sorts folder arrays by date.

Phase 2 (Compose/Reply)

- ✓ • Reply executes new EML generation with **In-Reply-To** threading.
- ✓ • Compose builds fresh EMLs.
- ✓ • Outputs natively save into the data room as encrypted files.